

Lecture Notes on NAND Flash Memory Technology

Chapter 3: Programming and Erasing Operations

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1 3.1 Programming Mechanisms

• Fowler-Nordheim Programming

Definition 1.1 (F-N Programming). Channel-to-floating gate electron injection through tunnel oxide:

- Applied voltage: $V_{\text{pgm}} = 15 - 20 \text{ V}$
- Tunnel field: $E_{\text{tun}} > 10 \text{ MV/cm}$
- Programming time: $\sim 100\text{-}500 \text{ s}$
- Mechanism: Direct tunneling

Programming current density:

$$J_{FN} = \frac{q^3 E_{ox}^2}{8\pi\hbar\phi_B} \exp\left(-\frac{8\pi\sqrt{2m^*}\phi_B^{3/2}}{3qhE_{ox}}\right) \quad (1)$$

Charge injection rate:

$$\frac{dQ}{dt} = J_{FN} \cdot A_{\text{tunnel}} \quad (2)$$

• Channel Hot Electron Programming

Alternative programming method:

$$E_{kin} = \frac{1}{2}mv^2 = q \cdot V_{ds} \cdot \alpha_{\text{impact}} \quad (3)$$

Hot electron injection efficiency:

$$\eta_{HE} = \frac{I_{gate}}{I_{drain}} \propto \exp\left(-\frac{q\phi_B}{qV_{ds}\alpha}\right) \quad (4)$$

2 3.2 Incremental Step Pulse Programming (ISPP)

• ISPP Algorithm

Theorem 2.1 (ISPP Principle). Precise V_{th} control through iterative programming:

$$V_{pgm}(n) = V_{start} + n \cdot \Delta V_{step} \quad (5)$$

$$\text{where } n = 0, 1, 2, \dots, N_{max} \quad (6)$$

Programming sequence:

1. Apply programming pulse: $V_{pgm}(n)$
2. Verify against target: V_{target}
3. If $V_{th} < V_{target}$: increment and repeat
4. If $V_{th} \geq V_{target}$: programming complete

• ISPP Parameters

Key parameters optimization:

$$t_{total} = N_{pulse} \cdot (t_{pulse} + t_{verify}) + t_{overhead} \quad (7)$$

Step size selection:

$$\Delta V_{step} = \frac{\sigma_{target}}{3} \quad (\text{for } 99.7\% \text{ yield}) \quad (8)$$

Programming efficiency:

$$\eta_{ISPP} = \frac{\Delta V_{thachieved}}{\Delta V_{thapplied}} = \alpha \cdot \eta_{injection} \quad (9)$$

• Multi-Level ISPP

For MLC/TLC/QLC programming:

$$V_{verify,i} = V_{read,i} + \text{margin} \quad (10)$$

Programming order strategies:

- **LSB $\rightarrow$ MSB**: Lower interference
- **All-level**: Faster but complex
- **Foggy-Fine**: Two-pass optimization

3 3.3 Erase Mechanisms

• Band-to-Band Tunneling Erase

Definition 3.1 (BTBT Erase). Hot hole injection mechanism:

- Apply positive voltage to substrate/body
- Generate hot holes at Si-SiO interface
- Inject holes to neutralize FG electrons

BTBT generation rate:

$$G_{BTBT} = A \cdot E^{2.5} \exp\left(-\frac{B}{E}\right) \quad (11)$$

Hot hole injection current:

$$I_{hole} = q \cdot G_{BTBT} \cdot V_{dep} \cdot A_{junction} \cdot \eta_{injection} \quad (12)$$

• Fowler-Nordheim Erase

Alternative erase mechanism:

$$J_{erase} = J_0 \exp\left(-\frac{B}{|E_{ox}|}\right) \quad (13)$$

Erase time constant:

$$\tau_{erase} = \frac{Q_{initial}}{J_{erase} \cdot A_{tunnel}} \quad (14)$$

4 3.4 Block Erase Operations

• Erase Algorithm

Block erase sequence:

1. Pre-program all cells to high V_{th} (optional)
2. Apply erase voltage: $V_{erase} = 15 - 20$ V
3. Verify all cells: $V_{th} < V_{erase,verify}$
4. Soft programming for over-erased cells

• Erase Optimization

Erase time modeling:

$$t_{erase} = t_0 + k \cdot \ln\left(\frac{Q_{initial}}{Q_{final}}\right) \quad (15)$$

Voltage optimization:

$$V_{opt} = V_{min} + \Delta V \cdot \sqrt{\frac{t_{spec}}{t_{natural}}} \quad (16)$$

5 3.5 Program/Erase Disturb Analysis

• Program Disturb

Unintended programming of unselected cells:

1. Self-Boosting Scheme

$$V_{channel} = V_{pass} - \Delta V_{couple} \quad (17)$$

Channel boosting efficiency:

$$\eta_{boost} = \frac{C_{ONO}}{C_{ONO} + C_{junction} + C_{depletion}} \quad (18)$$

2. Local Self-Boosting (LSB) Improved isolation:

$$V_{boost,LSB} = V_{boost,SB} \cdot \frac{N_{iso}}{N_{total}} \quad (19)$$

• Erase Disturb

Unwanted erase of unselected blocks:

$$\Delta V_{thdisturb} = \frac{Q_{loss}}{C_{total}} = f(V_{stress}, t_{stress}, T) \quad (20)$$

Disturb rate equation:

$$\frac{d(\Delta V_{th})}{dt} = A \exp\left(-\frac{E_a}{kT}\right) \cdot f(E_{field}) \quad (21)$$

6 3.6 Verify Operations

• Program Verify

Verification principles:

$$I_{cell}(V_{verify}) > I_{ref} \Rightarrow \text{PASS} \quad (22)$$

Verify margin:

$$V_{verify} = V_{read} + \text{margin} - \Delta V_{retention} - \Delta V_{disturb} \quad (23)$$

• Incremental Verify

Smart verify algorithm:

$$V_{verify}(n) = \begin{cases} V_{target} - 2\sigma & \text{if } n < N_{coarse} \\ V_{target} & \text{if } n \geq N_{coarse} \end{cases} \quad (24)$$

7 3.7 Margin Management

• Threshold Voltage Windows

Window budget allocation:

$$V_{window} = V_{spacing} + 2\sigma_{program} + \Delta V_{retention} + \Delta V_{disturb} \quad (25)$$

Statistical margin:

$$\text{Margin} = k_{\sigma} \cdot \sqrt{\sigma_{prog}^2 + \sigma_{retention}^2 + \sigma_{disturb}^2} \quad (26)$$

- **Adaptive Programming**

Dynamic step size adjustment:

$$\Delta V_{step}(n) = \Delta V_{initial} \cdot \exp\left(-\frac{n}{N_{decay}}\right) \quad (27)$$

Temperature compensation:

$$V_{pgm}(T) = V_{pgm}(25C) + TC \cdot (T - 25C) \quad (28)$$

8 Examples

Example 8.1 (ISPP Programming Time). Calculate total ISPP time for TLC programming: - Start voltage: 15V, Step: 0.5V, Max pulses: 20 - Pulse width: 20s, Verify time: 10s per level - 3 verify levels for TLC

Solution:

$$t_{pulse} = 20 \text{ s} \quad (29)$$

$$t_{verify} = 3 \times 10 = 30 \text{ s} \quad (30)$$

$$t_{cycle} = t_{pulse} + t_{verify} = 50 \text{ s} \quad (31)$$

$$t_{max} = 20 \times 50 = 1000 \text{ s} = 1 \text{ ms} \quad (32)$$

Average pulses needed: ~10 Average programming time: ~500 s

Example 8.2 (Disturb Rate Calculation). Calculate program disturb for: - Pass voltage: 10V, Programming voltage: 20V - Temperature: 85°C, Stress time: 1ms - Activation energy: 1.2 eV

Solution:

$$E_{tunnel} = \frac{10}{8 \times 10^{-9}} = 1.25 \times 10^9 \text{ V/m} \quad (33)$$

$$T = 85 + 273 = 358 \text{ K} \quad (34)$$

$$\text{Rate} = A \exp\left(-\frac{1.2 \times 1.6 \times 10^{-19}}{1.38 \times 10^{-23} \times 358}\right) \quad (35)$$

$$\approx A \exp(-38.8) \approx 10^{-17} A \quad (36)$$

Disturb is negligible for practical stress conditions.

9 Homework Problems

Assignment 3.1: Programming and Erase Operations

1. ISPP Optimization:

- Design ISPP for QLC (4 bits/cell)
- 16 threshold states, 6V total window
- Target distribution: = 150mV
- Optimize start voltage and step size

2. F-N Programming Analysis:

- Calculate programming time vs. voltage (15-22V)
- Tunnel oxide: 8nm, Temperature: 25°C, 85°C
- Include barrier lowering effects
- Plot current density and charge injection rate

3. Self-Boosting Design:

- 64-cell string with $V_{pass} = 10V$
- Calculate channel potential distribution
- Design for $\pm 10mV$ disturb per 10 cycles
- Compare SB vs. LSB schemes

4. Block Erase Modeling:

- BTBT erase with substrate bias
- Calculate erase time vs. voltage (12-20V)
- Include temperature dependence
- Design soft programming algorithm

5. Multi-Pass Programming:

- Implement Foggy-Fine programming
- Foggy pass: $\pm 1V$ accuracy
- Fine pass: $\pm 0.2V$ accuracy
- Analyze total programming time

6. Disturb Budget Analysis:

- 10 P/E cycles specification
- Program disturb: 10 stresses/cycle
- Read disturb: 10 reads/cycle
- Calculate margin allocation

7. Temperature Effects:

- Programming efficiency vs. T (-40°C to 85°C)
- Erase rate temperature dependence
- Design compensation algorithms
- Calculate activation energies

8. Advanced Verify Schemes:

- Smart verify with prediction
- Adaptive step size algorithm
- Multi-level concurrent verify

- Optimize for minimum program time

9. **3D NAND Programming:**

- String-to-string variation effects
- Layer-dependent programming
- Word line interference analysis
- Design compensation techniques

10. **Endurance Optimization:**

- Program/erase voltage scaling with cycles
- Adaptive pulse width adjustment
- Wear leveling impact on operations
- Design for 10 cycle specification

Next Chapter Preview: Chapter 4 will cover "Read Operations and Sensing" including sense amplifier design, read disturb mechanisms, and read retry optimization.